Notice that the value returned by the mathematical operator, $Im(\ )$, does not include the j. For example, $Im(3+4j)$ is equal to 4, not $4j$.

Complex numbers follow the same algebra as ordinary numbers, treating the quantity, j, as a constant. For instance, addition, subtraction, multiplication and division are given by:

EQUATION 30-1
Addition of complex numbers.

$$(a+bj)+(c+dj) = (a+c)+j(b+d)$$

EQUATION 30-2
Subtraction of complex numbers.

$$(a+bj)-(c+dj) = (a-c)+j(b-d)$$

EQUATION 30-3
Multiplication of complex numbers.

$$(a+bj)(c+dj) = (ac-bd)+j(bc+ad)$$

EQUATION 30-4
Division of complex numbers.

$$\frac{(a+bj)}{(c+dj)} = \left(\frac{ac+bd}{c^2+d^2}\right) + j\left(\frac{bc-ad}{c^2+d^2}\right)$$

Two tricks are used when manipulating equations such as these. First, whenever a j^2 term is encountered, it is replaced by -1. This follows from the definition of j, that is: $j^2 = (\sqrt{-1})^2 = -1$. The second trick is a way to eliminate the j term from the denominator of a fraction. For instance, the left side of Eq. 30-4 has a denominator of $c+dj$. This is handled by multiplying the numerator and denominator by the term $c-jd$, cancelling all the imaginary terms from the denominator. In the jargon of the field, switching the sign of the imaginary part of a complex number is called taking the **complex conjugate**. This is denoted by a star at the upper right corner of the variable. For example, if $Z = a+bj$, then $Z^* = a-bj$. In other words, Eq. 30-4 is derived by multiplying both the numerator and denominator by the complex conjugate of the denominator.

The following properties hold even when the variables A, B, and C are complex. These relations can be proven by breaking each variable into its real and imaginary parts and working out the algebra.

EQUATION 30-5
Commutative property.

$$AB = BA$$

EQUATION 30-6
Associative property.

$$(A+B)+C = A+(B+C)$$

EQUATION 30-7
Distributive property.

$$A(B+C) = AB+AC$$